

Governing AI’s Environmental Footprint: Gaps, Tradeoffs, and Recommendations for San Francisco

Focus area: AI infrastructure, energy policy, urban sustainability

Geography: San Francisco / California

Date: July 2026

EXECUTIVE SUMMARY

Artificial intelligence is expanding rapidly, and its reliance on energy-intensive data centers is generating measurable environmental and public health consequences, including greenhouse gas emissions, water consumption, and urban heat intensification. San Francisco’s position as a center of AI development makes it especially relevant to these policy questions, yet its 2026 Climate Action Plan does not account for AI-related emissions. Recent California legislation—including the vetoed SB 1047, the signed SB 53, and the pending SB 886 and SB 1168—reflects genuine legislative intent but contains structural limitations that undermine effectiveness. This brief identifies three core policy gaps and advances four targeted recommendations: standardizing environmental disclosure requirements, strengthening energy threshold mechanisms, accelerating the transition away from diesel backup infrastructure, and integrating AI growth projections into municipal climate planning.

Problem Statement

AI systems are powered by data centers: large, continuously operating facilities that consume substantial electricity and water. Global data center electricity consumption reached approximately 415 terawatt hours in 2025 and is projected to double by 2030, growing more than four times faster than any other sector. In 2023, U.S. data centers consumed around 17 billion gallons of water for cooling alone, a figure projected to increase nearly five-fold by 2028.

2x by 2030

Projected doubling of global data center electricity use within five years.

+1.5–2.4°C

Temperature increase documented near AI data center clusters within one year of operation.

Beyond energy and water, data centers contribute to the urban heat island (UHI) effect and release air pollutants through diesel backup generators—particulate matter, sulfur dioxide, nitrogen oxides, and volatile organic compounds—that are associated with cardiovascular disease in surrounding communities. These impacts are spatially concentrated: heat effects from data centers have been measured up to 4.5 kilometers away, with communities adjacent to data center clusters bearing a disproportionate share of the burden.

San Francisco's position as a center of AI investment makes it particularly exposed to these dynamics. Yet current governance frameworks—both at the city and state level—have not kept pace with the scale or specificity of AI's environmental footprint.

Current Policy Landscape

California has been among the most active states in attempting to regulate AI and data center growth. As of July 2026, there are four pieces of legislation central to this landscape:

SB 1047 — VETOED 2024 Would have required pre-deployment safety testing and detailed disclosure for large frontier AI models.
SB 53 — SIGNED INTO LAW 2025 A lighter version. Requires detailed security reporting before product release but does not address energy or environmental disclosure.
SB 886 — PENDING Would require large data centers to pre-fund fifteen-year contracts for zero-carbon energy infrastructure, shielding utility ratepayers from the financial burden of AI electricity demand.
SB 1168 — PENDING Would levy fines on data centers exceeding energy consumption thresholds, directing revenue toward residential electricity cost offsets.

San Francisco's 2026 Climate Action Plan outlines a pathway to net-zero emissions by 2040 and employs the GHG Protocol's three-scope framework to classify emissions sources. However, the plan does not reference AI infrastructure as a factor in the city's emissions trajectory. This is a significant omission given that AI energy consumption for cities and companies that rely on externally operated infrastructure falls under Scope 3 (indirect emissions across complex supply chains), the most difficult category to track and the one most likely to undermine net-zero targets if left unaddressed.

Identified Policy Gaps

Gap 1 — Disclosure without environmental scope. SB 53 mandates safety transparency but imposes no requirements on energy or environmental reporting. The absence of standardized metrics for electricity consumption, water usage, and GHG emissions means policymakers lack the data necessary to assess AI's actual environmental footprint.
Gap 2 — Exploitable energy thresholds. SB 886's 25-megawatt applicability threshold creates a structural loophole: companies can distribute energy loads across clusters of smaller, proximate facilities to remain below the threshold. This strategy not only avoids regulatory obligation but may amplify the UHI effect by concentrating heat-generating infrastructure in tighter geographic areas. Similarly, SB 1168's fine-based mechanism risks functioning as a predictable operating cost rather than a genuine consumption ceiling for large technology companies.
Gap 3 — Jurisdictional limits and planning omissions. Both pending energy bills apply only to infrastructure within California, leaving companies free to relocate data center operations to states with weaker oversight. At the municipal level, San Francisco's Climate Action Plan does not model AI-driven emissions growth, creating a structural gap between declared net-zero goals and the trajectory of the city's most rapidly expanding economic sector.

Recommendations

1 Establish standardized environmental disclosure requirements for AI developers and data center operators

California should extend existing AI transparency obligations to include mandatory reporting on electricity consumption, water usage, and GHG emissions, using a common methodology applicable across company sizes. This would enable cities like San Francisco to incorporate AI-related emissions into Scope 3 accounting within their climate plans and provide policymakers with the data necessary to set meaningful targets.

Tradeoff: Standardized reporting imposes compliance costs, which fall disproportionately on smaller operators. Tiered thresholds—scaling requirements by company size or model capability—could mitigate this without undermining the disclosure goal.

2 Amend SB 886's energy threshold to apply to aggregated loads across geographically proximate facilities

The current per-facility threshold should be restructured so that the combined energy demand of data centers operating within a defined geographic radius—for example, within a 7-kilometer zone, consistent with documented UHI impact ranges—is treated as a single unit for regulatory purposes. This would close the decentralization loophole and better align the bill's design with the spatial realities of heat and emissions impact.

Tradeoff: Aggregated thresholds introduce definitional and enforcement complexity. Clear geographic parameters and consistent metering standards would be necessary to make this workable.

3 Incentivize or require the transition from diesel backup generators to cleaner alternatives

Policymakers should establish timelines and financing mechanisms—such as tax credits, grant programs, or permitting incentives—for data centers to replace diesel backup infrastructure with battery storage systems or green hydrogen fuel cells. This is particularly urgent in communities adjacent to high concentrations of data centers, where generator emissions pose the most direct public health risk. SB 886's intent to shift financial burden away from ratepayers provides a useful model for structuring industry-funded transition programs.

Tradeoff: Alternative backup systems require significant upfront capital and must meet reliability standards under peak-demand and emergency conditions. Phased adoption timelines and reliability carve-outs for critical infrastructure periods would ease the transition.

4 Integrate AI infrastructure growth into San Francisco's Climate Action Plan and future municipal planning

The next revision of San Francisco's Climate Action Plan should explicitly model AI-driven Scope 3 emissions as a distinct and growing contributor to the city's emissions profile. This would allow the city to set sector-specific targets, identify mitigation strategies, and avoid the planning gap that currently exists between net-zero ambitions and AI-driven energy demand. Given the inherent uncertainty in AI growth projections, scenario-based modeling—including a high-growth case—would be more robust than a single-point forecast.

Tradeoff: AI's trajectory is genuinely difficult to predict, which complicates long-range planning. However, the absence of any planning assumption is more dangerous than an imperfect one. Scenario modeling explicitly accounts for uncertainty rather than ignoring it.

Conclusion

California's legislative activity on AI energy represents meaningful progress, but the structural limitations of current frameworks—in disclosure requirements, energy threshold design, jurisdictional reach, and climate planning—leave significant gaps between stated sustainability goals and the realities of AI's environmental footprint. San Francisco's position as a center of AI development makes it both uniquely exposed to these risks and uniquely positioned to lead in addressing them. The recommendations outlined here are not calls to constrain technological innovation, but to ensure that governance frameworks evolve alongside the technology they are designed to manage. The central policy question is not whether to regulate, but whether regulation can develop with sufficient precision and pace to remain meaningful.

Key Sources

International Energy Agency. (2025). Energy demand from AI.
<https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>

Leppert, R. (2025, October 24). What we know about energy use at U.S. data centers amid the AI boom. Pew Research Center.

Liu, R., Wu, Z., & Ren, S. (2025, November). An assessment of California data centers' environmental and public health impacts. Next 10.

Marinoni, A., et al. (2026). The data heat island effect: Quantifying the impact of AI data centers in a warming world. <https://doi.org/10.48550/ARXIV.2603.20897>

Redmond, M. (2026, March 27). San Francisco's climate change goals don't account for the AI boom. San Francisco Chronicle.

World Resources Institute & World Business Council for Sustainable Development. (2004). The greenhouse gas protocol. Greenhouse Gas Protocol.

This brief is adapted from my research paper, *AI Energy and Environmental Policy in San Francisco*, which includes full citations and analysis. This brief is intended as an educational and civic engagement resource.